Week 0: Assignment 0

Your last recorded submission was on 2024-04-23, 16:42 IST

1) What are the key features of python language? ***1 point***

- Object- oriented
- Concise and simple
- Both of the above
- None of the above

Yes, the answer is correct.
Score: 1
Accepted Answers:
Both of the above

2) State true or false ***1 point***
Statement: List in python is immutable but tupple is mutable

- True
- False

Yes, the answer is correct.
Score: 1
Accepted Answers:
False

3) For getting 3rd, 4th & 6th row of a datafile "df"in ***1 point***
Python programming, we can write:

- df.loc[[2,3,5]]
- df.loc[[3,4,5]]
- df.iloc[3,4,6]
- None of the above

Yes, the answer is correct.
Score: 1
Accepted Answers:
df.loc[[2,3,5]]

4) Bar Charts are used for : ***1 point***

- Continuous data
- Categorical data
- Both (a) & (b)
- None of the above

Yes, the answer is correct.
Score: 1
Accepted Answers:
Categorical data

Check Answers and Submit

Your score is: 4/4

Week 1: Assignment 1

The due date for submitting this assignment has passed.

Due on 2024-02-07, 23:59 IST.

Assignment submitted on 2024-02-06, 12:34 IST

1) Which of the following is never possible for variance? *1 point*

- Zero variance
- larger than the standard deviation
- Negative variance
- smaller than the standard deviation

Yes, the answer is correct.
Score: 1
Accepted Answers:
Negative variance

2) def m(data) *1 point*

Diff = max(data) – min(data)

return(Diff)

The above defined data function in Python programming, will calculate the?

- Inter quartile range
- mode
- median
- range

Yes, the answer is correct.
Score: 1
Accepted Answers:
range

3) Bar Charts are used for : *1 point*

- Continuous data
- Categorical data
- both a. and b.
- None of these

Yes, the answer is correct.
Score: 1
Accepted Answers:
Categorical data

4) Frequency polygons are used for: *1 point*

- Continuous data
- Categorical data
- both a. and b.
- None of the above

No, the answer is incorrect.
Score: 0
Accepted Answers:
Continuous data

5) m is an example of which of the following? *1 point*

- A population parameter
- sample statistic
- population variance
- mode

Yes, the answer is correct.
Score: 1
Accepted Answers:
A population parameter

6) Consider the following statements- *1 point*

Statement A : To "flatten" the dataframe, you can use the reset_index().

Statement B : Use the nunique() to get counts of unique values on a Pandas Series.

- Both statements are correct
- Both statements are false
- A is correct, B is false
- B is correct, A is false

Yes, the answer is correct.
Score: 1
Accepted Answers:
Both statements are correct

7) Which of the following statements in context of drawing an excellent graph is false: *1 point*

- The graph should not contain unnecessary adornments (sometimes referred to as chart junk)
- The scale on the vertical axis should begin at zero
- All axes should not be properly labelled
- The graph should contain a title

Yes, the answer is correct.
Score: 1
Accepted Answers:
All axes should not be properly labelled

8) Which of the following is not a measure of dispersion? *1 point*

- Skewness
- Kurtosis
- Range
- Percentile

Yes, the answer is correct.
Score: 1
Accepted Answers:
Percentile

9) Assume, you are given two lists: *1 point*
a = [1,2,3,4,5]
b = [6,7,8,9]

The task is to create a list which has all the elements of a and b in one dimension.
Output:
a = [1,2,3,4,5,6,7,8,9]

Which of the following option would you choose?

- a.append(b)
- a.extend(b)
- Any of the above
- None of these

Yes, the answer is correct.
Score: 1
Accepted Answers:
a.extend(b)

10)State the following true or false? *1 point*
Statement: Bimodal Data sets contains more than two modes

- True
- False

Yes, the answer is correct.
Score: 1
Accepted Answers:
False

Week 2: Assignment 2

The due date for submitting this assignment has passed.

Due on 2024-02-07, 23:59 IST.

Assignment submitted on 2024-02-07, 12:34 IST

1) Which of the following is each individual outcome of an experiment called? *1 point*

- the sample space
- a sample point
- an experiment
- an individual

Yes, the answer is correct.
Score: 1
Accepted Answers:
a sample point

2) A process that generates a well-defined outcomes is called as? *1 point*

- an event
- an experiment
- a sample point
- a sample space

Yes, the answer is correct.
Score: 1
Accepted Answers:
an experiment

3) Two events having nonzero probabilities *1 point*

- can be both mutually exclusive and independent
- cannot be both mutually exclusive and independent
- are always mutually exclusive
- are always independent

Yes, the answer is correct.
Score: 1
Accepted Answers:
cannot be both mutually exclusive and independent

4) A normal distribution with a standard deviation of 1 and mean of 0 is defined as? *1 point*

- a probability density function
- an ordinary normal curve
- a standard normal distribution
- None of these alternatives is correct

Yes, the answer is correct.
Score: 1
Accepted Answers:
a standard normal distribution

5) In a standard normal probability distribution, the area to the left portion of the mean is equal to? *1 point*

- -0.5
- 0.5
- any value between 0 and 1
- 1

Yes, the answer is correct.
Score: 1
Accepted Answers:
0.5

6) The weight of football players is normally distributed with a mean of 200 pounds and a standard deviation of 25 pounds. The probability of a player weighing more than 241.25 pounds is *1 point*

- 0.4505
- 0.0495
- 0.9505
- 0.9010

Yes, the answer is correct.
Score: 1
Accepted Answers:
0.0495

7) Refer to Q 6. The probability of a player weighing less than 250 pounds is *1 point*

- 0.4772
- 0.9772
- 0.0528
- 0.5

Yes, the answer is correct.
Score: 1
Accepted Answers:
0.9772

8) The Poisson probability distribution is which of the following type? *1 point*

- Continuous probability distribution
- Discrete probability distribution
- Uniform probability distribution
- Normal probability distribution

Yes, the answer is correct.
Score: 1
Accepted Answers:
Discrete probability distribution

9) Assuming a binomial experiment with p = 0.5 and a sample size of 100. The expected value of this distribution is? *1 point*

- 0.5
- 0.3
- 100
- 50

Yes, the answer is correct.
Score: 1
Accepted Answers:
50

10) The hypergeometric probability distribution is identical to *1 point*

- the Poisson probability distribution
- the binomial probability distribution
- the normal distribution
- None of these alternatives is correct

Yes, the answer is correct.
Score: 1
Accepted Answers:
None of these alternatives is correct

Week 3: Assignment 3

The due date for submitting this assignment has passed.

Due on 2024-02-14, 23:59 IST.

Assignment submitted on 2024-02-13, 12:12 IST

1) Why one should not go for sampling? *1 point*

- It is less costly to administer than a census
- The person authorising the study is comfortable with the sample.
- Because the research process is sometimes destructive
- None of the above

Yes, the answer is correct.
Score: 1
Accepted Answers:
None of the above

2) Stratified random sampling is a method of selecting a sample in which *1 point*

- the sample is first divided into strata
- various strata are selected from the sample
- The population is first divided into strata, and then random samples are drawn from each stratum
- None of the above

Yes, the answer is correct.
Score: 1
Accepted Answers:
The population is first divided into strata, and then random samples are drawn from each stratum

3) If the true proportion of customers who are below 20 years is P=0.35, what is the probability that a sample size of 100 yields a sample proportion between 0.3 to 0.4 *1 point*

- 0.961
- 0.827
- 0.706
- 0.53

No, the answer is incorrect.
Score: 0
Accepted Answers:
0.706

4) Calculate the probability of getting 12 heads in 20 attempts from a fair coin. *1 point*

- 0.120
- 0.240
- 0.280
- 0.301

Yes, the answer is correct.
Score: 1
Accepted Answers:
0.120

5) A question paper contains 90 multiple-choice questions. Each correct answer carries 1 mark. There are 4 alternative answers (A, B, C or D), of which only one is correct. Mr X answers these questions randomly (i.e. without preparation). What is the probability that X gets at least 10 marks? *1 point*

- 0.9997
- 0.7894
- 0
- 0.001

Yes, the answer is correct.
Score: 1
Accepted Answers:
0.9997

6) On average, 5 % of items supplied by manufacturer X are defective. If a batch of 10 items is inspected: what is the probability that 2 items are defective *1 point*

- 0.065
- 0.075
- 0.085
- 0.095

Yes, the answer is correct.
Score: 1
Accepted Answers:
0.075

7) A car distributor in City Y experiences an average of 2.5 car sales daily. Find the probability that on a randomly selected day, they will sell 5 car: *1 point*

- 0.0668
- 0.544
- 0.082
- 0.205

Yes, the answer is correct.
Score: 1
Accepted Answers:
0.0668

8) In question 7, Find the probability that on a randomly selected day, they will sell no cars:: *1 point*

- 0.0668
- 0.544
- 0.082
- 0.205

Yes, the answer is correct.
Score: 1
Accepted Answers:
0.082

9) In question 7, Find the probability that on a randomly selected day, they will sell at most 2 cars: *1 point*

- 0.0668
- 0.544
- 0.082
- 0.205

Yes, the answer is correct.
Score: 1
Accepted Answers:
0.544

10)In question 7, Find the probability that on a randomly selected day, they will sell exactly one car: *1 point*

- 0.0668
- 0.544
- 0.082
- 0.205

Yes, the answer is correct.
Score: 1
Accepted Answers:
0.205

Week 4: Assignment 4

The due date for submitting this assignment has passed.

Due on 2024-02-21, 23:59 IST.

Assignment submitted on 2024-02-21, 23:45 IST

1) Null hypothesis, $H_0: m_1 - m_2 = 0$ *1 point*

- Upper tail test
- Lower tail test
- Two tail test
- F Test

Yes, the answer is correct.
Score: 1

Accepted Answers:
Two tail test

2) If we have a sample size of 15 and the population standard deviation is known, we will use: *1 point*

- t- test for hypothesis testing
- z-test for hypothesis testing
- both t and z test
- F Test

No, the answer is incorrect.
Score: 0

Accepted Answers:
z-test for hypothesis testing

3) The quality-control manager at a Li-BATTERY factory needs to determine whether the mean life of a large shipment of Li-Battery is equal to the specified value of 375 hours. The process standard deviation is known to be 100 hours. A random sample of 64 batteries indicates a sample mean life of 350 hours. State the null hypotheses. *1 point*

- Mu = 375
- Mu ≤ 375
- Mu = 350
- Mu ≥ 350

Yes, the answer is correct.
Score: 1

Accepted Answers:
Mu = 375

4) In question 3, At the alpha = 0.05 level of significance, is there any evidence that the mean life differs from 375 hours? *1 point*

- Yes, there is
- No, there is not
- None of the above

Yes, the answer is correct.
Score: 1

Accepted Answers:
Yes, there is

5) In question 3, the computed p-value is: *1 point*

- 0.0456
- 0.456
- 0.0228
- 0.228

Yes, the answer is correct.
Score: 1

Accepted Answers:
0.0456

6) In question 3, at a 95% confidence interval, the estimate of the population mean life of the battery is: *1 point*

- 325.5 to 379.5
- 325.5 to 374.5
- 320.5 to 379.5
- 320.5 to 374.5

Yes, the answer is correct.
Score: 1

Accepted Answers:
325.5 to 374.5

7) The mean cost of a hotel room in a city is $168 per night. A random sample of 25 hotels resulted in X-bar = $172.50 and sample standard deviation s = 15.40. Calculate the t statistic. *1 point*

- 2
- -2
- 1.46
- -1.46

Yes, the answer is correct.
Score: 1

Accepted Answers:
1.46

8) At the = 0.05 level. In question 7, the decision on the null hypothesis is: *1 point*

- Do not reject
- reject
- None of the above

Yes, the answer is correct.
Score: 1

Accepted Answers:
Do not reject

9) In the hypothesis testing procedure refers to *1 point*

- 1 - the level of significance
- the critical value
- the confidence level
- level of significance

Yes, the answer is correct.
Score: 1

Accepted Answers:
level of significance

10) If a hypothesis test leads to the rejection of the null hypothesis, *1 point*

- a Type II error must have been committed
- a Type II error may have been committed
- a Type I error must have been committed
- a Type I error may have been committed

No, the answer is incorrect.
Score: 0

Accepted Answers:
a Type I error may have been committed

Week 5: Assignment 5

The due date for submitting this assignment has passed.

Due on 2024-02-28, 23:59 IST.

Assignment submitted on 2024-02-27, 15:47 IST

1) In the analysis of variance procedure (ANOVA), the term "factor" refers to: ***1 point***

- the dependent variable
- the independent variable
- different levels of treatment
- the critical value of F

Yes, the answer is correct.
Score: 1
Accepted Answers:
the independent variable

2) In a problem of ANOVA involving 3 treatments and 10 observations per treatment, SSE = 399.6. The MSE for this situation is ***1 point***

- 130.2
- 48.8
- 14.8
- 30.0

Yes, the answer is correct.
Score: 1
Accepted Answers:
14.8

3) The 'F' ratio in a completely randomised ANOVA is the ratio of ***1 point***

- MSTR/MSE
- MST/MSE
- MSE/MSTR
- MSE/MST

No, the answer is incorrect.
Score: 0
Accepted Answers:
MSTR/MSE

4) An ANOVA procedure is applied to data obtained from 6 samples where each sample contains 10 observations. The degrees of freedom for the critical value of F are ***1 point***

- 6 numerator and 20 denominator degrees of freedom
- 5 numerator and 20 denominator degrees of freedom
- 5 numerator and 54 denominator degrees of freedom
- 6 numerator and 20 denominator degrees of freedom

Yes, the answer is correct.
Score: 1
Accepted Answers:
5 numerator and 54 denominator degrees of freedom

5) In an ANOVA problem if SST = 120 and SSTR = 80, then SSE is ***1 point***

- 20
- 40
- 80
- 120

Yes, the answer is correct.
Score: 1
Accepted Answers:
40

6) The critical F value with 8 numerator and 29 denominator degrees of freedom at alpha = 0.01 is ***1 point***

- 2.18
- 3.20
- 3.53
- 3.95

No, the answer is incorrect.
Score: 0
Accepted Answers:
3.20

7) Two Independent simple random samples are taken to test the difference between the means of two populations. The standard deviations are not known, but are assumed to be equal. The sample sizes are n1 = 15 and n2 = 35. The correct distribution to use is the ***1 point***

- t distribution with 51 degrees of freedom
- z distribution with 50 degrees of freedom
- z distribution with 49 degrees of freedom
- t distribution with 48 degrees of freedom

Yes, the answer is correct.
Score: 1
Accepted Answers:
t distribution with 48 degrees of freedom

8) The sampling distribution of two populations $\bar{p}_1 - \bar{p}_2$ is approximated by a ***1 point***

- t-distribution with n1 + n2 degrees of freedom
- normal distribution
- t-distribution with n1 + n2 - 1 degrees of freedom
- t-distribution with n1 + n2 + 2 degrees of freedom

Yes, the answer is correct.
Score: 1
Accepted Answers:
normal distribution

9) Mean marks obtained by male and female students of schools ABCD in the first unit test are shown below. ***1 point***

	Male	Female
Sample Size	64	36
Sample Mean Marks	44	41
Population Variance (σ^2)	128	72

The standard error for the difference between the two means is

- 4.0
- 7.46
- 4.24
- 2.0

Yes, the answer is correct.
Score: 1
Accepted Answers:
2.0

10)Refer to Q. 9: If you are interested in testing whether or not the average marks of males is significantly greater than that of females, the test statistic is ***1 point***

- 2
- 1.5
- 1.96
- 1.645

Yes, the answer is correct.
Score: 1
Accepted Answers:
1.5

Week 6: Assignment 6

The due date for submitting this assignment has passed.

Due on 2024-03-06, 23:59 IST.

Assignment submitted on 2024-03-06, 22:00 IST

1) The model developed from sample data having the form of $\hat{y} = b_0 + b_1 x$ is known as *1 point*

- regression equation
- correlation equation
- estimated regression equation
- regression model

Yes, the answer is correct.
Score: 1
Accepted Answers:
estimated regression equation

2) In regression analysis, which of the following is not a required assumption about the error term ?? *1 point*

- The expected value of the error term is one
- The variance of the error term is the same for all values of X
- The values of the error term are independent
- The error term is normally distributed

Yes, the answer is correct.
Score: 1
Accepted Answers:
The expected value of the error term is one

3) A regression analysis between sales (Y in $1000) and advertising (X in dollars) resulted in the following equation *1 point*

Y= 30,000 + 5 X

The above equation implies that an

- increase of $5 in advertising is associated with an increase of $5,000 in sales
- increase of $1 in advertising is associated with an increase of $5 in sales
- increase of $1 in advertising is associated with an increase of $35,000 in sales
- increase of $1 in advertising is associated with an increase of $5,000 in sales

Yes, the answer is correct.
Score: 1
Accepted Answers:
increase of $1 in advertising is associated with an increase of $5,000 in sales

4) In a regression and correlation analysis if r squared = 1, then *1 point*

- SSE = SST
- SSE = 1
- SSR = SSE
- SSR = SST

Yes, the answer is correct.
Score: 1
Accepted Answers:
SSR = SST

5) SSE can never be *1 point*

- larger than SST
- smaller than SST
- equal to 1
- equal to zero

Yes, the answer is correct.
Score: 1
Accepted Answers:
larger than SST

6) For the given data determine the R-squared value *1 point*

Data:

Miles travel	Petrol Consumption in litre
20	1
45	3
56	5
34	2
28	1.6
49	3.7

- 0.887
- 0.956
- 0.945
- 0.932

Yes, the answer is correct.
Score: 1
Accepted Answers:
0.956

7) In question no. 6 , when testing the hypothesis of slope, we will: *1 point*

- Accept the null hypothesis
- Reject the null hypothesis
- Can't state any conclusion
- None of the above

Yes, the answer is correct.
Score: 1
Accepted Answers:
Reject the null hypothesis

8) In Question 6, determine a 95% confidence interval for B1 to test the hypotheses *1 point*

- (0.045, 0.138)
- (0.055, 0.148)
- (0.065, 0.158)
- (0.075, 0.138)

Yes, the answer is correct.
Score: 1
Accepted Answers:
(0.075, 0.138)

9) State TRUE or FALSE – *1 point*

Statement: The variance of error is same for all values of the independent variable

- True
- False

Yes, the answer is correct.
Score: 1
Accepted Answers:
True

10)Which of the following is possible for the coefficient of determination: *1 point*

- It can be larger than 1
- It is less than one
- It can be less than -1
- None of these alternatives is correct

Yes, the answer is correct.
Score: 1
Accepted Answers:
It is less than one

Week 7: Assignment 7

The due date for submitting this assignment has passed.

Due on 2024-03-13, 23:59 IST.

Assignment submitted on 2024-03-12, 12:31 IST

1) The interval estimate of the mean value of y (dependent variable) for a given value of x is defined as? **1 point**

- Prediction interval estimate
- Confidence interval estimate
- Average regression
- X vs Y correlation interval

No, the answer is incorrect.
Score: 0
Accepted Answers:
Confidence interval estimate

2) Which of the following is called Standard Error? **1 point**

- T-statistic squared
- Square root of SSE
- Square root of SST
- Square root of MSE

Yes, the answer is correct.
Score: 1
Accepted Answers:
Square root of MSE

3) Which of the following is true about multiple regression model? **1 point**

- It has only one independent variable
- It has more than one dependent variable
- It has more than one independent variable
- It has at least 2 dependent variable

Yes, the answer is correct.
Score: 1
Accepted Answers:
It has more than one independent variable

4) In a multiple regression model, the error term ε is assumed to **1 point**

- Have a mean of 1
- Have a variance of 0
- Have a standard deviation of 1
- Be normally distributed

Yes, the answer is correct.
Score: 1
Accepted Answers:
Be normally distributed

5) For a multiple regression model with 2 independent variables, R.sq = 0.904 and adjusted R. sq = 0.88, determine the number of observations (n) **0 points**

- 6
- 7
- 9
- 10

No, the answer is incorrect.
Score: 0
Accepted Answers:
10

6) If the R.sq value is small for a model with a large number of independent variables, the adjusted coefficient of determination _____________ **1 point**

- Must be positive
- Can be negative
- Is zero
- Can't say

Yes, the answer is correct.
Score: 1
Accepted Answers:
Can be negative

7) Which one of the statements is true regarding residuals in regression analysis? **1 point**

- Mean of residuals is always 0
- Mean of residuals is always < 0
- Mean of residuals is always > 0
- There is no such rule for residuals

Yes, the answer is correct.
Score: 1
Accepted Answers:
Mean of residuals is always 0

8) In a simple linear regression model (one independent variable), if we increase the input variable by 1 unit, how much will the output variable change? **1 point**

- By 1
- No change
- By its slope
- None of these

Yes, the answer is correct.
Score: 1
Accepted Answers:
By its slope

9) If linear regression model has train error = 0, then test error ________ **1 point**

- Is also 0
- Is non-zero
- Is always equal to train error
- Cannot comment on Test error

No, the answer is incorrect.
Score: 0
Accepted Answers:
Cannot comment on Test error

10)Which of the following evaluation metrics is used to evaluate a model while modelling a continuous output variable? **1 point**

- AUC-ROC
- Accuracy
- Logloss
- Mean-Squared-Error

Yes, the answer is correct.
Score: 1
Accepted Answers:
Mean-Squared-Error

Week 8: Assignment 8

The due date for submitting this assignment has passed.

Due on 2024-03-20, 23:59 IST.

Assignment submitted on 2024-03-20, 23:01 IST

1) For categorical data with 'n' categories, the number of dummy variables will be________ ***1 point***

- n
- n-1
- n+1
- 2n

Yes, the answer is correct.
Score: 1

Accepted Answers:
n-1

2) In estimation of regression parameters ***1 point***

- The likelihood function is a function of only σ
- The values of $\beta 0$ .. βn and σ should be such that, they maximizes the likelihood function.
- Both (a) and (b)
- none of the above

Yes, the answer is correct.
Score: 1

Accepted Answers:
The values of $\beta 0$ .. βn and σ should be such that, they maximizes the likelihood function.

3) In logistic regression, the null hypothesis tested is: ***1 point***

- H0: β = 0
- H0: β ≠ 0
- H0: μ = 0
- H0: μ ≠ 0

Yes, the answer is correct.
Score: 1

Accepted Answers:
H0: β = 0

4) In logistic regression, ***1 point***

- The graph doesn't follow S shape curve
- The dependent variable is categorical
- The estimated value of dependent variable is not probability
- None of the above.

Yes, the answer is correct.
Score: 1

Accepted Answers:
The dependent variable is categorical

5) State true or false: G statistic is used to check the individual significance of the independent variables ***1 point***

- True
- False

Yes, the answer is correct.
Score: 1

Accepted Answers:
False

6) What is the range of values as output from the sigmoid function? ***1 point***

- 0 to 1
- -1 to -1
- -1 to 0
- None of these

Yes, the answer is correct.
Score: 1

Accepted Answers:
0 to 1

7) State True or False: The method of least squares is used to predict the population parameters with any probability distribution. ***1 point***

- True
- False

Yes, the answer is correct.
Score: 1

Accepted Answers:
False

8) Suppose you have been given a fair coin and you want to find out the odds of getting heads. Which of the following option is true for such a case? ***1 point***

- Odds will be 0
- Odds will be 0.5
- Odds will be 1
- None of these

Yes, the answer is correct.
Score: 1

Accepted Answers:
Odds will be 1

9) Large values of the log-likelihood statistic indicate: ***1 point***

- That there are a greater number of explained vs. unexplained observations.
- That the statistical model fits the data well.
- That as the predictor variable increases, the likelihood of the outcome occurring decreases.
- That the statistical model is a poor fit for the data.

Yes, the answer is correct.
Score: 1

Accepted Answers:
That the statistical model fits the data well.

10)The logit function(given as l(x)) is the log of odds function. What could be the range of logit function in the domain x = [0,1]? ***1 point***

- $(-\infty, \infty)$
- (0,1)
- $(0, \infty)$
- $(-\infty, 0)$

Yes, the answer is correct.
Score: 1

Accepted Answers:
$(-\infty, \infty)$

Week 9: Assignment 9

The due date for submitting this assignment has passed.

Due on 2024-03-27, 23:59 IST.

Assignment submitted on 2024-03-27, 20:43 IST

1) State true or false: Statement: there is no difference between, E(y) = 0 + 1x and y = 0 + 1x + e , both are regression equations *0 points*

- True
- False

No, the answer is incorrect.
Score: 0
Accepted Answers:
False

2) Which of the following statements is correct: *1 point*

- Sensitivity in ROC analysis is called True Positive Rate(tpr)
- Specificity in ROC analysis is not called True Negative Rate (tnr)
- Specificity in ROC analysis is called True Positive Rate(tpr)
- Sensitivity in ROC analysis is called True Negative Rate (tnr)

Yes, the answer is correct.
Score: 1
Accepted Answers:
Sensitivity in ROC analysis is called True Positive Rate(tpr)

3) In ROC analysis when the Threshold value is Higher: *1 point*

- Specificity decreases
- Sensitivity decreases
- Both a. and b.
- None of the above

Yes, the answer is correct.
Score: 1
Accepted Answers:
Sensitivity decreases

4) Sensitivity in ROC analysis is defined as: *1 point*

(Here, TP : True Positive, TN: True Negative, FP: False Positive, FN: False Negative)

- FP/ (FP + TN)
- FN / (TP + FN)
- TN/ (TN + FP)
- TP / (TP + FN)

Yes, the answer is correct.
Score: 1
Accepted Answers:
TP / (TP + FN)

5) In ROC analysis, a classifier is called 'good' if it has _____ *1 point*

- Low TPR and Low FPR
- Low TPR and High FPR
- High TPR and Low FPR
- High TPR and High FPR

Yes, the answer is correct.
Score: 1
Accepted Answers:
High TPR and Low FPR

6) For the given confusion matrix, compute the recall *1 point*

	True Positive	True Negative
Predicted Positive	8	3
Predicted Negative	2	7

- 0.73
- 0.7
- 0.78
- 0.8

Yes, the answer is correct.
Score: 1
Accepted Answers:
0.8

7) State true or False: Precision is inversely proportional to recall *1 point*

- True
- False

Yes, the answer is correct.
Score: 1
Accepted Answers:
False

8) State True or False: Standardization of features is not required before training a Logistic regression model *1 point*

- True
- False

Yes, the answer is correct.
Score: 1
Accepted Answers:
True

9) State True or False: Standardization of features is not required before training a Logistic regression model *0 points*

- Linear Regression errors values have to be normally distributed but in the case of Logistic Regression it is not the case
- Logistic Regression errors values have to be normally distributed but in the case of Linear Regression it is not the case
- Both Linear Regression and Logistic Regression error values have to be normally distributed
- Both Linear Regression and Logistic Regression error values have not to be normally distributed

Yes, the answer is correct.
Score: 0
Accepted Answers:
Linear Regression errors values have to be normally distributed but in the case of Logistic Regression it is not the case

10) Which of the following is true regarding the logistic function for any value "x"? *1 point*

A. Logistic(x): is a logistic function of any number "x"
B. Logit(x): is a logit function of any number "x"
C. Logit_inv(x): is an inverse logit function of any number "x"

- Logistic(x) = Logit(x)
- Logistic(x) = Logit_inv(x)
- Logit_inv(x) = Logit(x)
- None of these

No, the answer is incorrect.
Score: 0
Accepted Answers:
Logistic(x) = Logit_inv(x)

Week 10: Assignment 10

The due date for submitting this assignment has passed.

Due on 2024-04-03, 23:59

Assignment submitted on 2024-04-03, 12:42 IST

1) Sampling distribution for a goodness of fit test is the — *1 point*

- Poisson distribution
- t distribution
- normal distribution
- chi-square distribution

Yes, the answer is correct.
Score: 1
Accepted Answers:
chi-square distribution

2) Goodness of fit test is always conducted as a — *1 point*

- lower-tail test
- upper-tail test
- middle test
- None of these alternatives is correct.

No, the answer is incorrect.
Score: 0
Accepted Answers:
upper-tail test

3) State True or False: Statement: Null hypothesis for chi square test of independence assumes that, all the proportions are equal. — *1 point*

- True
- False

No, the answer is incorrect.
Score: 0
Accepted Answers:
True

4) Statistical test conducted to determine whether to reject or not reject a hypothesized probability distribution for a population is known as a _______ — *1 point*

- contingency test
- probability test
- goodness of fit test
- None of these alternatives is correct

Yes, the answer is correct.
Score: 1
Accepted Answers:
goodness of fit test

5) What is the minimum no. of variables/ features required to perform clustering? — *1 point*

- 0
- 1
- 2
- 3

Yes, the answer is correct.
Score: 1
Accepted Answers:
1

6) Which of the following method is used for finding optimal clusters in K-Mean algorithm? — *1 point*

- Elbow method
- Manhattan method
- Euclidian method
- None of these

Yes, the answer is correct.
Score: 1
Accepted Answers:
Elbow method

7) Movie Recommendation systems are an example of_____ — *1 point*

1. Classification 2. Clustering 3. Reinforcement Learning 4. Regression

- 2 Only
- 1 and 2
- 1 and 3
- 1, 2, 3 and 4

No, the answer is incorrect.
Score: 0
Accepted Answers:
1, 2, 3 and 4

8) How can Clustering (Unsupervised Learning) be used to improve the accuracy of the Linear Regression model (Supervised Learning): — *1 point*

1. Creating different models for different cluster groups.

2. Creating an input feature for cluster ids as an ordinal variable.

3. Creating an input feature for cluster centroids as a continuous variable.

4. Creating an input feature for cluster size as a continuous variable

- 1 Only
- 1 and 2
- 1 and 4
- 1, 2, 3 and 4

No, the answer is incorrect.
Score: 0
Accepted Answers:
1, 2, 3 and 4

9) Let x_1 = (1,2) and x_2 = (3,5) be the co-ordiantes for two objects. The Euclidean and Manhattan distance between these two objects is ________ respectively — *1 point*

- 4.2 and 3
- 3.15 and 2
- 3.61 and 5
- None of these

Yes, the answer is correct.
Score: 1
Accepted Answers:
3.61 and 5

10) State true or false: Discriminant Analysis does not require the grouping variable to be known at the beginning — *1 point*

- True
- False

Yes, the answer is correct.
Score: 1
Accepted Answers:
False

Week 11: Assignment 11

The due date for submitting this assignment has passed.

Due on 2024-04-10, 23:59 IST.

Assignment submitted on 2024-04-10, 23:06 IST

1) Which library is used for calculating distance measures in clustering using python? *0 points*

- distance_matrix
- scipy.spatial
- scipy_spatial
- distance.matrix

No, the answer is incorrect.
Score: 0
Accepted Answers:
distance_matrix

2) Formula for dissimilarity computation between two objects for categorical variables is – *1 point*

Here p is a categorical variable and m denotes the number of matches.

- D(i, j) = p-m / p
- D(i, j) = p-m / m
- D(i, j) = m - p / p
- D(i, j) = m - p / m

Yes, the answer is correct.
Score: 1
Accepted Answers:
D(i, j) = p-m / p

3) Select the correct option for a data set with 7 objects and an interval-scaled variable 'f' we have the following measurements: *1 point*
f = (1, 2, 3, 4, 5, 8, 50) containing one outlying value.

- Std deviation (std_f) and mean absolute deviation (s_f) are having the same effect of the outlier.
- Mean absolute deviation (s_f) is more affected by the outlier
- Std deviation (std_f) is less affected by the outlier
- Std deviation(std_f) is more affected by the outlier.

No, the answer is incorrect.
Score: 0
Accepted Answers:
Std deviation(std_f) is more affected by the outlier.

4) Select the correct statement about the standardization in the following options *1 point*

- Standardizing the data always gives inefficient result while making clusters
- Standardizing the data always beneficial during clustering analysis
- The variables having an absolute value may not efficient after standardization during clustering analysis
- Outliers can not be detected by standardized data

No, the answer is incorrect.
Score: 0
Accepted Answers:
The variables having an absolute value may not efficient after standardization during clustering analysis

5) Which of the following can act as possible termination conditions in K-Means? *1 point*

1. For a fixed number of iterations.
2. Assignment of observations to clusters does not change between iterations. Except for cases with a bad local minimum.
3. Centroids do not change between successive iterations.
4. Terminate when RSS falls below a threshold.

- 1, 3 and 4
- 1,2,3 and 4
- 2 and 3
- None of these

Yes, the answer is correct.
Score: 1
Accepted Answers:
1,2,3 and 4

6) In the figure below, if you draw a horizontal line on y-axis for y=2. What will be the number of clusters formed? *1 point*

https://drive.google.com/file/d/18pFcGnHQgsDTpHfeeWA09NaRWrRI-2_v/view?usp=sharing

- 1
- 2
- 3
- 4

Yes, the answer is correct.
Score: 1
Accepted Answers:
2

7) Which of the following clustering requires merging approach? *1 point*

- Partitional
- Naive Bayes
- Hierarchical
- None of the above

Yes, the answer is correct.
Score: 1
Accepted Answers:
Hierarchical

8) State True or False: Hierarchical clustering should primarily be used for exploration *1 point*

- True
- False

Yes, the answer is correct.
Score: 1
Accepted Answers:
True

9) State True or False: For finding dissimilarity between two clusters in hierarchical clustering, average-link is the only metric used *1 point*

- True
- False

Yes, the answer is correct.
Score: 1
Accepted Answers:
False

10)If two variables V1 and V2, are used for clustering. Which of the following are true for K means clustering with k =3? *1 point*

1. If V1 and V2 has a correlation of 1, the cluster centroids will be in a straight line
2. If V1 and V2 has a correlation of 0, the cluster centroids will be in straight line

- 1 only
- 2 only
- 1 and 2
- None of the above

Yes, the answer is correct.
Score: 1
Accepted Answers:
1 only

Week 12: Assignment 12

The due date for submitting this assignment has passed.

Due on 2024-04-17, 23:59 IST.

Assignment submitted on 2024-04-17, 22:02 IST

1) Which clustering algorithm works well when the shape of the clusters is hyper-spherical? *1 point*

- K means
- Agglomerative Hierarchical clustering
- Divisive Hierarchical clustering
- All of the above

Yes, the answer is correct.
Score: 1
Accepted Answers:
K means

2) In decision tree, an internal node represents – *1 point*

- A test on an attribute
- An outcome of the test
- Entire sample population
- Holds a class label

Yes, the answer is correct.
Score: 1
Accepted Answers:
A test on an attribute

3) Choose the correct statement about the CART model *1 point*

- CART is an unsupervised learning technique
- CART is a supervised learning technique
- CART adopts a greedy approach
- Both b. and c.

Yes, the answer is correct.
Score: 1
Accepted Answers:
Both b. and c.

4) Which library is used to built the decision tree model- *1 point*

- Decision tree classifier
- DecisionTreeClassifier
- Decision_Tree_Classifier
- Decision_tree_model

Yes, the answer is correct.
Score: 1
Accepted Answers:
DecisionTreeClassifier

5) State True or False: Gini Index enforces the resulting tree to have multiway splits *1 point*

- True
- False

Yes, the answer is correct.
Score: 1
Accepted Answers:
False

6) Chance nodes are represented by ___________ *1 point*

- Disks
- Squares
- Circles
- Triangles

Yes, the answer is correct.
Score: 1
Accepted Answers:
Circles

7) _______is the measure of uncertainty of a random variable, it characterizes the impurity of an arbitrary collection of examples. *1 point*

- Information Gain
- Gini Index
- Entropy
- None of the above

Yes, the answer is correct.
Score: 1
Accepted Answers:
Entropy

8) End Nodes are represented by __________ *1 point*

- Disks
- Squares
- Circles
- Triangles

No, the answer is incorrect.
Score: 0
Accepted Answers:
Triangles

9) Decision tree learners may create biased trees if some classes dominate. What's the solution of it? *1 point*

- Balance the dataset prior to fitting
- Imbalance the dataset prior to fitting
- Balance the dataset after fitting
- None of the above

Yes, the answer is correct.
Score: 1
Accepted Answers:
Balance the dataset prior to fitting

10)Suppose, your target variable is the price of a house using Decision Tree. What type of tree do you need to predict the target variable? *1 point*

- Classification tree
- Regression Tree
- Clustering tree
- Dimensionality reduction tree

Yes, the answer is correct.
Score: 1
Accepted Answers:
Regression Tree